



Mahindra University, Hyderabad

École Centrale School of Engineering

Minor I (2024 - Batch)

Program: B. Tech.

Branch: CM

Year: II

Semester: II

Subject: Statistical Methods (MA2220)

Date: 25/02/2026

Start Time: 02:00 PM

Time Duration: 1.5 Hours

Max. Marks: 30

Instructions:

- 1) Each question carries 10 marks.
- 2) All questions are compulsory.
- 3) Please start each answer on a separate page and ensure that you clearly number the responses.
- 4) It is essential to explain each step. Correct outcomes without any description will not be evaluated.

The data is collected from office employees to understand their **daily commuting patterns, work habits, and wellness indicators**. The objective is to explore how commuting and lifestyle factors relate to employee productivity and wellbeing.

- **ID:** Unique employee identification number.
- **Age:** Age of the employee in completed years.
- **Mode:** Primary mode of transport used to commute (Car/Bus/Bike/Metro).
- **Dept:** Work department of the employee.
- **C-time:** Average one-way commute time (in hours).
- **Break:** Average break time during work (in hours).
- **Drinks:** Number of energy drinks consumed per day.
- **Level:** Self-reported workload category with ordered levels Low < Moderate < Heavy.
- **Productivity Score:** Weekly productivity score on a 10-point scale.
- **Remote Option:** Whether the employee has remote work flexibility (Yes/No).

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ID	Age	Mode	Dept	C-time	Break	Drinks	Level	Productivity Score	Remote Option
1	24	Bus	HR	0.30	0.4	1	Moderate	7.1	Yes
2	29	Car	Tech	0.55	0.6	2	Heavy	8.4	No
3	31	Bike	Sales	0.18	0.3	1	Low	6.8	Yes
4	27	Metro	Tech	0.72	0.8	3	Heavy	9.0	No
5	35	Car	Finance	1.60	1.5	4	Heavy	9.5	No
6	22	Bus	HR	0.42	0.5	1	Moderate	7.3	Yes
7	30	Bike	Sales	0.12	0.2	0	Low	6.4	Yes
8	28	Metro	Tech	0.95	1.0	3	Moderate	8.7	No
9	33	Car	Finance	0.63	0.7	2	Moderate	8.1	No
10	26	Bus	HR	0.25	0.4	1	Low	7.0	Yes
11	37	Car	Tech	2.10	1.8	5	Heavy	9.6	No
12	25	Metro	Sales	0.48	0.6	2	Moderate	7.8	Yes
13	32	Bike	Finance	0.08	0.2	0	Low	6.6	Yes
14	29	Car	Tech	1.25	1.2	4	Heavy	9.2	No
15	34	Metro	HR	0.80	0.9	2	Moderate	8.3	No

Q 1:

10 marks

- (i) The dataset above contains multiple types of variables. Read the description of each variable carefully and answer the following.
- (a) Identify all Quantitative and Qualitative variables. [2 marks]
- (b) Among the qualitative variables, classify each as Nominal or Ordinal. [2 marks]
- (ii) Create side-by-side boxplots of Productivity Score grouped by Remote Option and briefly compare the medians, variability, and any potential outliers observed in these two groups. [5 marks]
- (iii) Suppose you are part of a decision making committee reviewing the remote work policy, state whether you would recommend continuing or revoking the Remote Option and justify your decision using statistical reasoning based on the plot. [1 marks]

Q 2:

10 marks

- (i) Let X denote the variable Drinks, representing the number of energy drinks consumed per day by an employee. Assume that $X \sim \text{Binomial}(7, p)$, where p denotes the probability that an employee consumes an energy drink in a single opportunity (trial) during the day, and each employee has a maximum of 7 such opportunities.
- (a) Using the Drinks data provided in the table, compute the Maximum Likelihood Estimate of p , $\hat{p}_{MLE}$. [3 marks]

P.T.O.

- (b) Suppose the office has 150 employees. Using $\hat{p}_{MLE}$, determine the expected (average) total number of drinks required in the office per day. [2 marks]
- (ii) Let Y denote the variable C-time, representing the one-way commute time (in hours) of an employee. Assume that Y follows Exponential distribution with rate parameter $\lambda > 0$.
- (a) Using the C-time data provided in the table, compute Maximum Likelihood Estimate of λ , $\hat{\lambda}_{MLE}$. [3 marks]
- (b) Using your estimate, determine the probability that an employee's commute time exceeds the sample mean (computed from C-time data). [2 marks]
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Q 3:

10 marks

- (i) Let $X_1, X_2, \dots, X_{10}$ be a random sample from $U(0, 1)$. Let

$$X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(10)}$$

denote the order statistics.

- (a) What is the probability density function of the fourth order statistic $X_{(4)}$. [2 marks]
- (b) Find $\mathbb{E}(X_{(10)})$. [3 marks]
- (ii) A Linear Congruential Method (LCM) is defined by

$$X_{n+1} = (17X_n + 5) \bmod 37, \quad X_0 = 3.$$

- (a) Generate the first 6 pseudo-random numbers $X_1, X_2, \dots, X_6$ and obtain the corresponding uniform random numbers $U_1, U_2, \dots, U_6$. [2 marks]
- (b) Using the uniform random number pairs $\{(U_1, U_2), (U_3, U_4), (U_5, U_6)\}$ obtained in part (a) and the Box-Muller transformation, generate 3 standard normal random numbers. [3 marks]
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